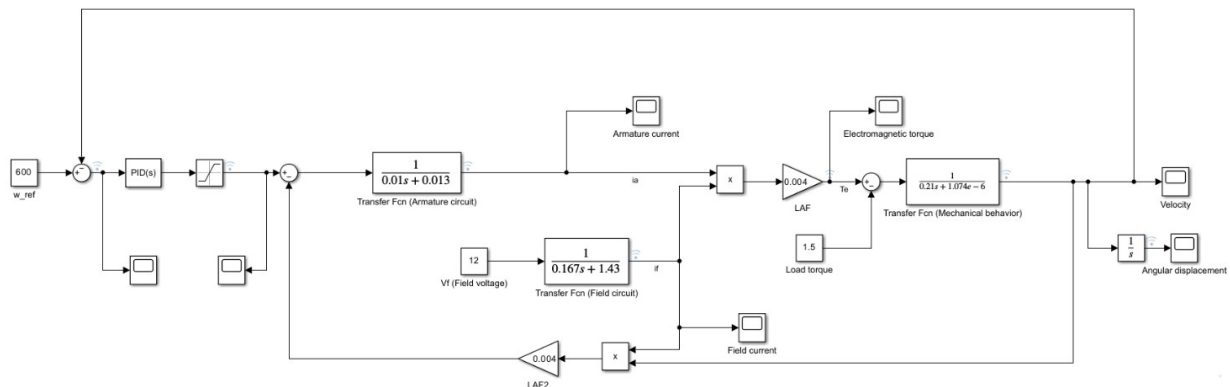


Modeling and Speed Control of a Separately Excited DC Motor

A concise English portfolio report



Closed-loop Simulink model with PI control and armature-voltage saturation

1.5 N m

Load torque

600 rad/s

Reference speed

Kp = 0.035

PI proportional gain

Ki = 0.015

PI integral gain

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Original work completed as a three-person academic team project.

1. Project Overview

The project develops a dynamic model of a separately excited DC motor under a constant mechanical load. The open-loop response is first evaluated, then a PI controller is added to regulate shaft speed. Both configurations are implemented in MATLAB/Simulink and simulated for 20 seconds.

Design objectives

- Select a motor configuration suitable for a constant-torque load.
- Build electrical, electromagnetic, and mechanical subsystems from first principles.
- Compare analytical steady-state values with simulation results.
- Regulate speed at 600 rad/s while limiting armature voltage to 0-24 V.

Operating conditions and parameters

Parameter	Value
Motor configuration	Separately excited DC motor
Load	Constant torque, $T_L = 1.5 \text{ N}\cdot\text{m}$
Armature supply	$V_a = 24 \text{ V}$ in open loop; 0-24 V in closed loop
Field supply	$V_f = 12 \text{ V}$
Armature circuit	$r_a = 0.013 \text{ }\Omega$, $L_{AA} = 0.01 \text{ H}$
Field circuit	$r_f = 1.43 \text{ }\Omega$, $L_{FF} = 0.167 \text{ H}$
Mechanical subsystem	$J = 0.21 \text{ kg}\cdot\text{m}^2$, $B_m = 1.074 \times 10^{-6} \text{ N}\cdot\text{m}\cdot\text{s}/\text{rad}$
Electromechanical coupling	$L_{AF} = 0.004 \text{ H}$

Modeling assumptions

The model uses lumped, linear electrical and mechanical parameters. Magnetic saturation, brush voltage drop, converter switching, thermal effects, and parameter variation are not represented. The load torque is constant.

2. Mathematical Model

The model is assembled from the field circuit, armature circuit, torque relation, and rotor dynamics.

Field and armature equations

$$\begin{aligned}V_f &= r_f i_f + L_{FF} \frac{di_f}{dt} \\V_a &= r_a i_a + L_{AA} \frac{di_a}{dt} + e_b \\e_b &= L_{AF} i_f \omega\end{aligned}$$

Electromagnetic torque and rotor dynamics

$$\begin{aligned}T_e &= L_{AF} i_f i_a = K i_a \\J \frac{d\omega}{dt} &= T_e - B_m \omega - T_L\end{aligned}$$

With a constant field supply, the steady field current is approximately $12/1.43 = 8.39$ A. The effective motor constant is therefore $K = 0.004 \times 8.39 = 0.03356$.

Open-loop steady-state estimate

$$\begin{aligned}\omega_{ss} &= \frac{V_a - r_a(T_L/K)}{K} \\ \omega_{ss} &\approx 697.7 \text{ rad/s}\end{aligned}$$

The simulated open-loop steady-state speed is also about 697.7 rad/s, which provides a direct consistency check between the analytical model and the Simulink implementation.

Control structure

The PI controller uses the difference between the 600 rad/s reference and the measured shaft speed. Its output commands the armature voltage, followed by a saturation block that enforces the 0-24 V drive limit.

$$C(s) = K_p + \frac{K_i}{s}, \quad K_p = 0.035, \quad K_i = 0.015$$

3. Simulink Implementation

Two models are provided in the repository. The first evaluates the motor with fixed armature and field voltages. The second closes the speed loop with a PI controller and voltage saturation.

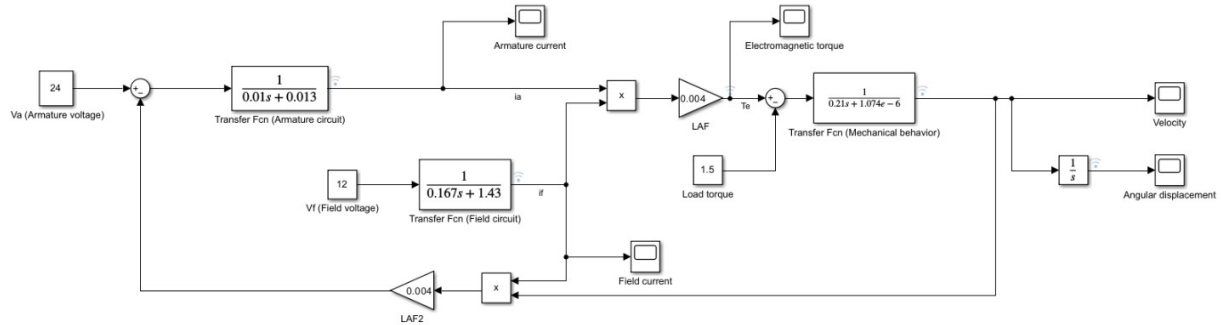


Figure 1. Open-loop model - repository file: *dc_motor_open_loop.slx*

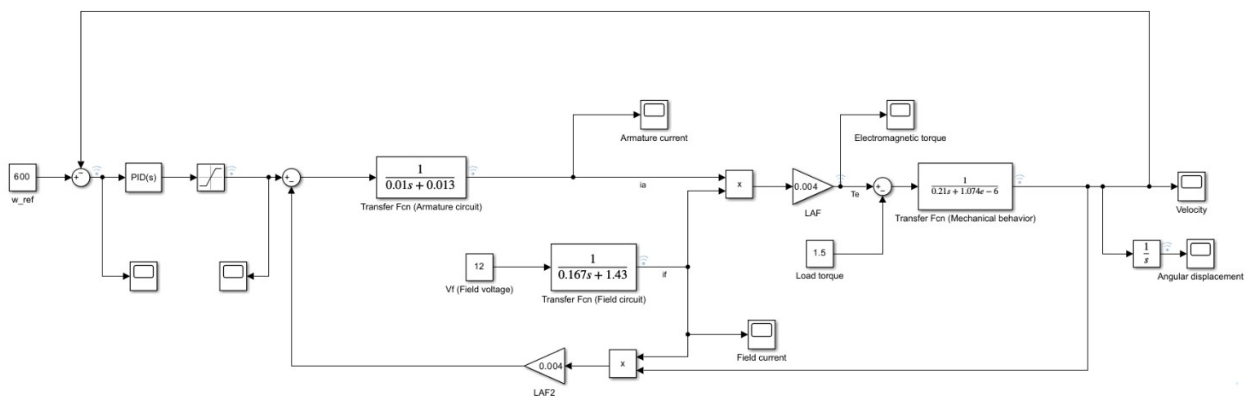


Figure 2. Closed-loop PI model - repository file: *dc_motor_pi_control.slx*

Signal flow

- The field circuit produces field current from the constant 12 V supply.
- Back EMF is subtracted from the commanded armature voltage.
- Armature current produces electromagnetic torque through the motor constant K.
- Load torque and viscous damping oppose acceleration in the mechanical subsystem.

Simulation setup

Simulation time is 20 s. The models were created in MATLAB/Simulink R2023b and store their numerical parameters inside the model files; no external MATLAB initialization script is required.

4. Open-Loop Response

With $V_a = 24$ V, $V_f = 12$ V, and $T_L = 1.5$ N m, the motor accelerates from rest and settles near 697.7 rad/s. This is above the desired 600 rad/s operating point, so a feedback controller is required.

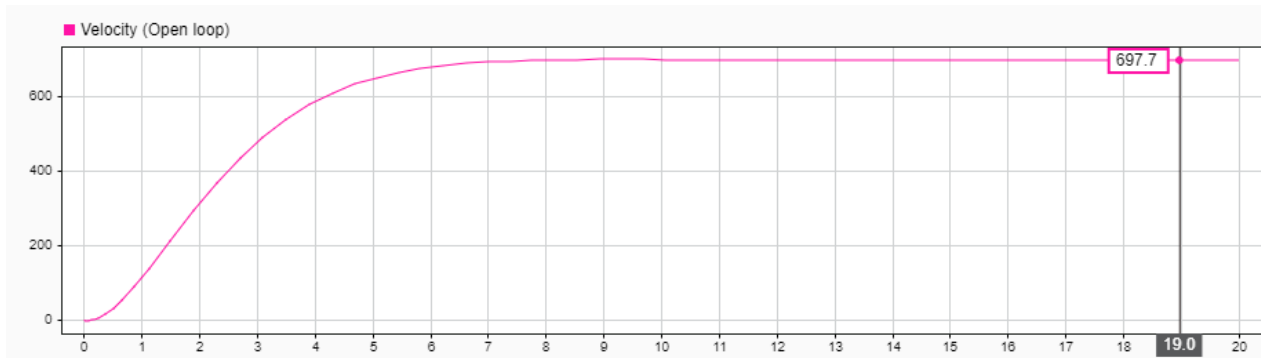


Figure 3. Open-loop speed response; steady-state speed is approximately 697.7 rad/s

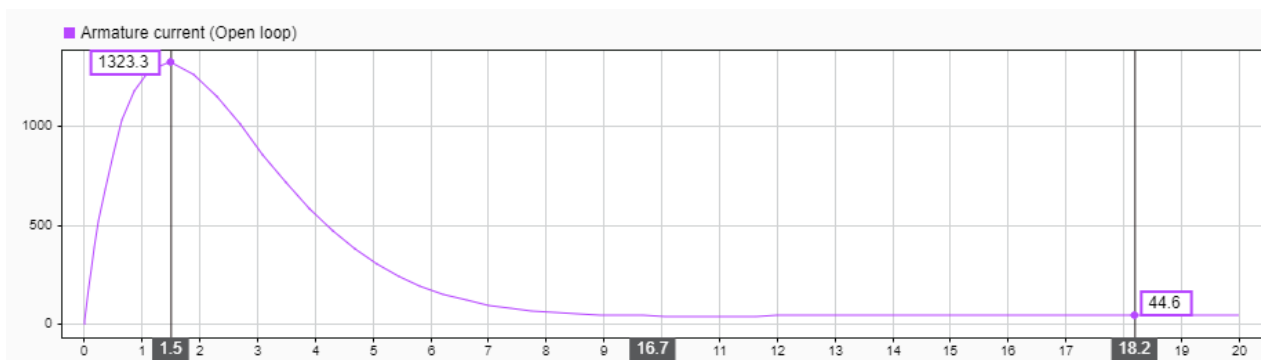


Figure 4. Open-loop armature current; the large startup transient decays to about 44.6 A

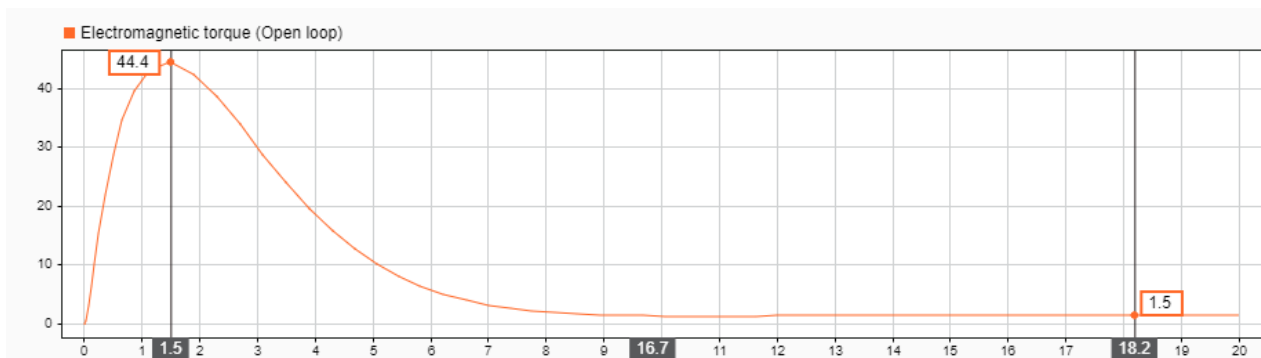


Figure 5. Open-loop electromagnetic torque; steady-state torque approaches the applied load

Interpretation

The response confirms the expected DC motor behavior: high current and torque during acceleration, followed by lower current once back EMF develops. The model reaches the analytical speed estimate, but open-loop operation cannot enforce the target speed when voltage and load are fixed.

5. PI Speed Control and Results

A proportional-only trial with $K_p = 0.03$ produced a large steady-state error. Adding integral action removed the offset. The final controller uses $K_p = 0.035$ and $K_i = 0.015$, with the armature command limited to 24 V.

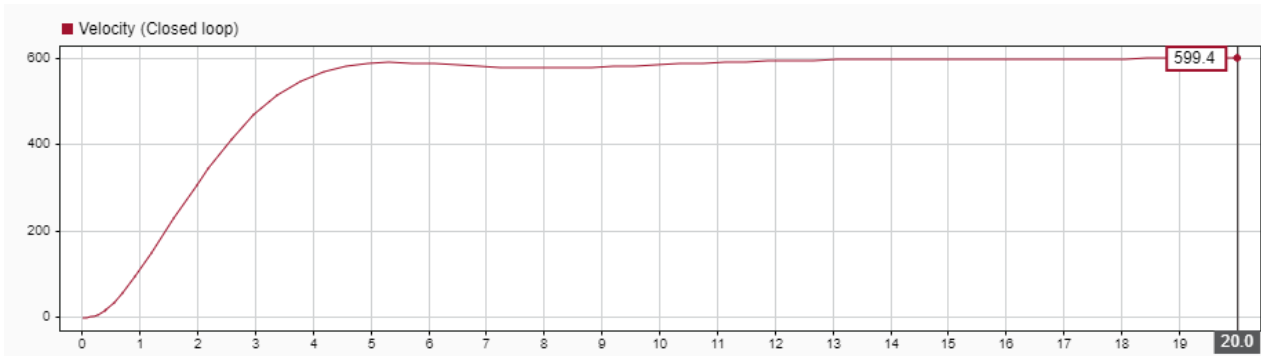


Figure 6. Closed-loop speed response; final speed is approximately 599.4 rad/s

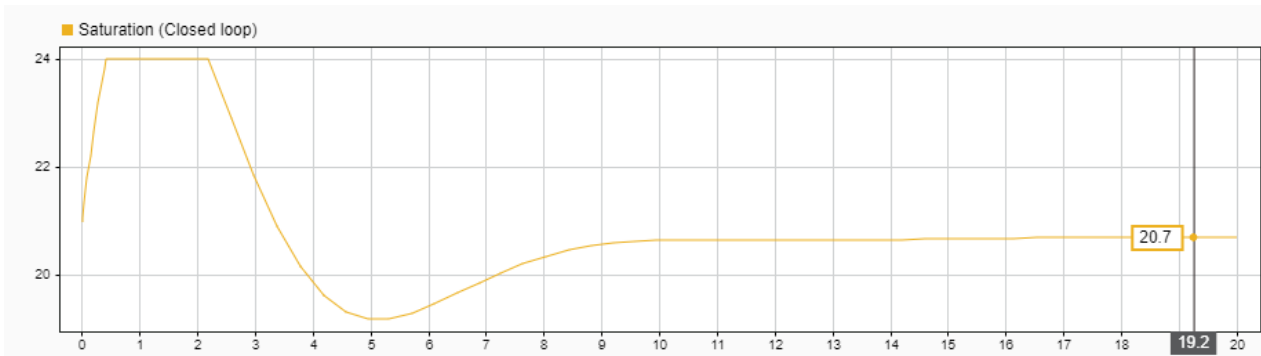


Figure 7. Controller output and armature voltage; steady-state value is about 20.7 V

Metric	Open loop	PI closed loop
Steady-state speed	697.7 rad/s	599.4 rad/s
Reference speed	Not applied	600 rad/s
Armature voltage	24 V constant	~20.7 V steady state
Speed error vs. 600 rad/s	+97.7 rad/s	~-0.6 rad/s

Engineering assessment

The PI controller brings the motor close to the 600 rad/s reference and removes almost all steady-state error. The simulation also predicts a very large startup current, above 1.3 kA. A practical drive would require current limiting, converter dynamics, protection, and experimental validation before hardware use.

Conclusion and next steps

The project demonstrates a consistent physics-based motor model and a basic closed-loop speed controller. Useful extensions include an inner current loop, anti-windup, load-step testing, parameter sensitivity, and comparison with measured motor data.

Selected references from the original report: Ogata, Modern Control Engineering (2009); Krause et al., Analysis of Electric Machinery and Drive Systems (2013); Krishnan, Electric Motor Drives (2001).